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#include <stdio.h>

#define MAX 5

int queue[MAX];

int front = -1, rear = -1;

void insert(int value) {
    if ((front == 0 && rear == MAX - 1) || (front == rear + 1)) {
        printf("Queue Overflow! Cannot insert %d\n", value);
    } else {
        if (front == -1) {
            front = rear = 0;
        } else if (rear == MAX - 1 && front != 0) {
            rear = 0;
        } else {
            rear = rear + 1;
        }
        queue[rear] = value;
        printf("%d inserted into queue.\n", value);
    }
}

void delete() {
    if (front == -1) {
        printf("Queue Underflow! Cannot delete.\n");
    } else {
        printf("%d deleted from queue.\n", queue[front]);
        if (front == rear) {
            front = rear = -1;
        } else {
            front = (front + 1) % MAX;
        }
    }
}

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    }
}

void display() {
    if (front == -1) {
        printf("Queue is empty.\n");
    } else {
        printf("Queue elements: ");
        int i = front;
        while (1) {
            printf("%d ", queue[i]);
            if (i == rear) break;
            i = (i + 1) % MAX;
        }
        printf("\n");
    }
}

int main() {
    int choice, value;
    while (1) {
        printf("\n--- Circular Queue Menu ---\n");
        printf("1. Insert\n2. Delete\n3. Display\n4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);
        switch (choice) {
            case 1:
                printf("Enter value to insert: ");
                scanf("%d", &value);
                insert(value);
                break;

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        case 2:
            delete();
            break;
        case 3:
            display();
            break;
        case 4:
            printf("Exiting program.\n");
            return 0;
        default:
            printf("Invalid choice! Try again.\n");
    }
}
}

```

Output

```

--- Circular Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 1
1 inserted into queue.

--- Circular Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 2
2 inserted into queue.

--- Circular Queue Menu ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value to insert: 3
3 inserted into queue.

```

--- Circular Queue Menu ---

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 1

Enter value to insert: 4

4 inserted into queue.

--- Circular Queue Menu ---

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 2

1 deleted from queue.

--- Circular Queue Menu ---

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 3

Queue elements: 2 3 4

--- Circular Queue Menu ---

1. Insert
2. Delete
3. Display
4. Exit

Enter your choice: 4

Exiting program.

Process returned 0 (0x0) execution time : 35.735 s